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Identifying intelligent data utilization in bioprocesses: overview of current research activities, opportunities and barriers.

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Abstract

Biological transformation, amplified by digital transformation, has seen a surge in interest in recent years. Efforts to develop sustainable technologies for a sustainable economy are reinforcing the increasing convergence between technology and biology. This will inevitably lead to disruptive technologies that will drive established technologies out of the market. For this reason, it is important to examine current research activities in the field of data usage in bioprocesses and bioproduction. By analyzing current research activities in bioproduction, this study will facilitate future research, identify research gaps and thus help to make out challenges in the development of biointelligent systems.

Keywords: Biointelligent production; Bioproduction; Data usage; Algorithm; Sensor

1. Introduction

Sustainability has emerged as an increasingly critical consideration in production, prompting a growing interest in biotransformed or biointelligent production [1, 2]. The shift towards bio-hybrid, circular production of goods is considered by many to be one of the most promising approaches to comprehensively restructure existing production patterns [3]. This emerging trend of biointelligent production is characterized by the use of biological principles, living organisms, and components to sustainably produce goods, generate new properties, or functions [3]. The organic component employed in biological systems or processes is referred to as bio-ware. The convergence of bio-ware, hardware, and software has resulted in biointelligent systems (BIS), which are intelligent systems that emulate the self-regulation and self-optimization mechanisms of natural organisms, resulting in increased efficiency and adaptability [3]. BIS face a significant challenge in effectively monitoring the biological component as biological systems are complex and highly unpredictable. This issue is particularly evident in bioprocesses, which are increasingly relevant in various fields, including healthcare, food production, and biotechnology. To enhance the efficiency and sustainability of bioprocesses, there has been a surge in interest in developing intelligent systems capable of collecting and utilizing data to monitor and control biological systems [4]. By systematically analyzing the process data, intelligent systems can assist in optimizing the process and even acquire knowledge from the data.

The extraction and utilization of data obtained from biological systems pose a fundamental question in bioproduction [4]. This paper endeavors to examine the

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existing literature on data and data formats that are collected during bioproduction processes. The study is divided into two key components: the first component explores literature that elucidates technologies used to retrieve data from biological systems. The second component categorizes the various data formats that are generated during the bioproduction process. We have mapped the literature to identify research trends and have discussed possible future scenarios and provided a future research agenda. By presenting a comprehensive review of the current state-of-the-art of data collection in bioproduction, this study will facilitate future research and help to identify challenges in the development of biointelligent systems.

2. Methodology

In this study, we conducted a systematic search to identify relevant literature on data in biological productions within the field of engineering science, chemical science, and biological science. The search strategy was tailored to the Scopus database, and the variants of the following search terms were used: "bioprocess", "biointelligence", "bioinspiration", "biological transformation," "biologicalization," "bio-convergence", "biorevolution", "bioproduction," "bioengineering", "bioproduction", "biofoundry" and "biodigital," in combination with "production", "industry", and "data". The search encompassed journal articles, review papers, and research reports published in English only from database inception until 2023 and excluded variants of the keywords human, mammals, articles, and plants. Further, the research excluded the subject areas arts, social science, and psychology.

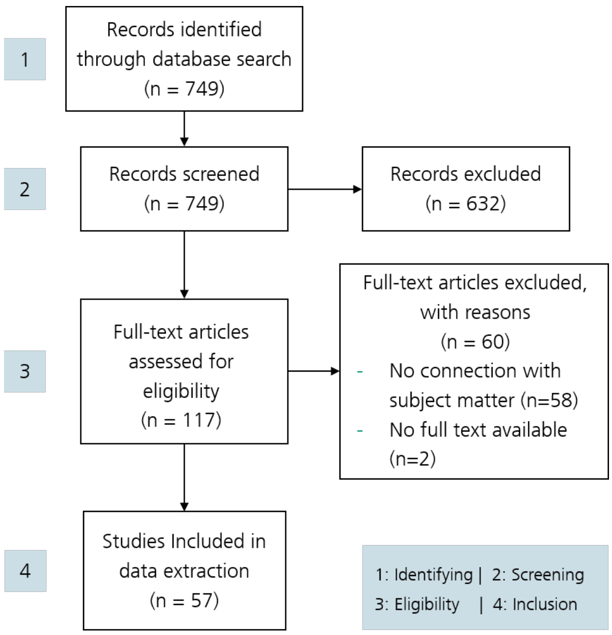

Fig. 1.The PRISMA methodology used in this study.

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement guided the selection criteria, which followed the Identification, Screening, Eligibility, and Inclusion steps [5]. The search spanned from 2003-2023, with all articles before 2003 excluded from the search. At the screening stage, 632 records were excluded, and 117 were extracted. Only original research articles, review papers, conference papers, and book chapters were included in the study. The abstracts of the articles were deeply examined to analyze and refine the articles, ensuring the quality and relevance of academic literature included in the review process. At a later stage, each research paper was carefully evaluated, and after the exclusion of studies written in non-English languages or studies with no connection to the subject matter, 42 articles were selected following the inclusion and exclusion criteria.

3. Results and Interpretations

3.1. Descriptive Analysis

Over the last 20 years, there has been a steady increase in the number of publications in the field of bioproduction. This is evident in Fig. 1, which includes publications up to 2022. While this suggests growing interest in the field, the relatively low number of works indicates that it is still a niche area of research. This is likely due to the fact that bioproduction is not yet as widely adopted as conventional production methods, partly because biological systems are inherently variable and can be challenging to control consistently. Bioproduction also requires specialized knowledge and skills, as well as significant initial investments in research and development, equipment, and infrastructure. Despite these challenges, bioproduction has great potential for sustainability and efficiency, and it is an area of growing interest in the scientific community.

Fig. 2.The production of publications by year from 2003 – 2022.

Fig. 3 shows the ten countries with the highest number of corresponding authors. Germany is at the top with eleven authors, India and the USA have seven authors each, followed by China and the UK with six. The remaining five countries have four or three corresponding authors. These results show that Germany is the most active in this research area, which is reflected in the highest number of authors. Furthermore, the graph illustrates Europe's strong interest in this area, as five of the top ten countries and a total of 27 of the 53 corresponding authors in the graph are from Europe.

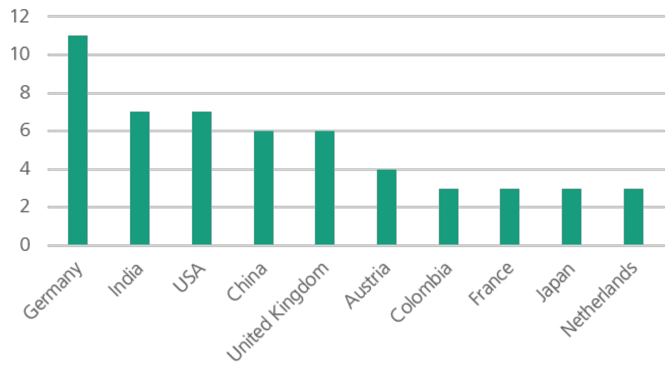

Fig. 3. Corresponding Author's Countries.

The researches with the highest citation count and therefore the biggest resonance are from Patil et al. [6], which proposes a logical data representation framework for electricity-driven bioproduction processes and Claßen et al. [7], which discusses the use of spectroscopic sensors for in-line bioprocess monitoring in research and pharmaceutical industrial applications.

To get greater insights of the themes of the investigated studies the following sections will discuss the hardware and software of the technologies, which where the focus of the research papers and the opportunities and challenges arising from them.

3.2. Monitoring Technologies

Efficient process control is essential for improving bioprocess productivity. Process analytics tools (PAT) as part of Quality by Design (QbD) help maintain consistent process performance and product quality by identifying critical quality attributes (CQAs) and corresponding critical process parameters (CPPs) that must be monitored and controlled [8]. PAT enables the timely measurement of CQAs and CPPs to ensure consistent product quality. The relationship between CQAs and CPPs must be understood to enable online monitoring of bioprocesses for the safe and profitable production of high-quality products. Detecting errors at an early stage is necessary to take corrective action before they impact production. Efficient process control requires a sensor to detect changes, a monitoring recipe to determine actions, control logic to control process variables, and verification to ensure the effectiveness of manipulations [8, 9].

Three kind of sensor systems can generally be used to monitor bioproduction processes, namely inline, at-line and offline sensors. Inline sensors provide continuous data without the need for sampling, while at-line sensors require continuous sampling and transport to an analytical system for further analysis. Offline analysis, on the other hand, requires individual samples to be taken from the process and analyzed in a laboratory, which can lead to significant time delays. Typically, physical parameters such as temperature, liquid level and pressure are monitored with in-line sensors, while biological variables such as cell viability are still monitored offline or, in the best case, at-line measurements [10]. Becker et al. provide further insights on this topic [10].

Recent advances in biochemistry, photonics and information technology have encouraged the development of powerful chemical-optical sensors that can monitor bioprocesses using various physical properties and constants of newly developed materials and the biological components they measure. These advances offer a number of advantages, such as miniaturization, flexibility, embedded sensing, fast at-line sensing, multi-sensing and continuous quantitative or qualitative measurements. Spectroscopic PAT tools [9] for non-invasive screening of cell culture media, optical density measurement [7], fluorometry [11], spectral methods such as near-infrared [12] and nuclear magnetic resonance-based metabolomics [13] are some of the examples. Entire systems consisting of multiple sensors such as the Respiratory Activity Monitoring System (RAMOS) [14], and GE Healthcare's Biacore Surface Plasmon Resonance system [15] are other examples of sensors and systems that enable near real-time concentration measurements of bioprocess components.

A new type of sensor are biosensors, which are a powerful tool in bioprocess monitoring, utilizing a biological sensing element and transducer to detect and convert biological signals into electronic output. The sensing element can be an enzyme, antibody, or other biological molecule that binds to specific target molecules in a sample, generating a signal that is detected and converted into an electronic output. The work of Full et al. exemplifies this in an overview of current odor biosensors [16].

Advanced algorithms and access to large amounts of process data have made monitoring solutions and process control systems increasingly powerful and comprehensive. In the bioprocess industry, advanced control strategies such as AI-based conventional control strategies and advanced control systems such as fuzzy logic, model predictive control, adaptive control and neural network-based control are being used to develop robust and powerful models for predictive operational control [9]. These strategies are particularly important for controlling bioreactors in which cells respond dynamically to changes in the surrounding culture media [9]. Wavelet transform, Hölder exponent evaluation and watershed segmentation have been proposed for physiological state detection based on bioreactor sensor signal segmentation [17, 18]. The aim of this method is to improve existing models and optimize bioreactor performance.

Approaches that do not aim to optimize the bioreactor but to improve the bioprocess itself are exemplified by the work of Simutis and Lübbert, with the combination of unscented Kalman filters (UKF) and support vector regression (SVR) modelling to improve the monitoring and optimization of a process for the production of recombinant therapeutic proteins [19]. UKF is used to estimate the process variables, while SVR is used to model the relationship between different parameters to optimize the process. To provide an efficient and accurate method for estimating parameters of nonlinear systems, several algorithms are used, including local search methods, simulated annealing, population heuristics and differential evolution [20]. Independent component analysis is used as a fingerprinting tool to identify differences between developing batches [21–23]. Generalized additive models [24], and hierarchical clustering [21] are data-based methods for monitoring errors in fermentation, which are important for reactor safety and maintaining product quality. Neural networks are used to identify effective gene deletions and determine the best gene deletion models for xylitol production and growth yield in E. coli bacteria, and to estimate complex states of biological processes in the presence of perturbations using a combination of machine learning and mathematical modelling techniques [25, 26]. These approaches and techniques have been shown to improve the accuracy, efficiency and productivity of bioprocess monitoring and optimization [27]. Decision support tools for bioproduction optimization include deterministic and stochastic optimization models, scheduling and capacity planning tools, and software such as GE Healthcare's Unicorn system control software [15]. These tools link biological and chemical processes to business risks and financial trade-offs, aiming to reduce costs and lead times. They use advanced algorithms and data analytics techniques to provide integrated knowledge for planning and controlling bioproduction operations. By analyzing results and implementing new algorithms, these tools enable bioprocess engineers to make informed decisions and optimize production and therefore crucial for improving efficiency and productivity in the bioprocess industry [9, 28, 29].

The combination of hardware and software for indirect measurement of parameters in bioprocesses leads to the development of so-called soft sensors. The high complexity of the general characteristics and variability of bioprocesses and the lack of reliable online sensors for most process variables lead to costly and inefficient offline or at-line assessments. For this reason, soft sensors are used as potential tools for real-time determination of biomass, product, metabolites, amino acid concentration and other CQAs. Soft sensors have been shown to be useful for batch analysis and monitoring, using multivariate statistical process control or process performance monitoring methods based on multivariate data analysis [21, 23]. They play an important role in effective monitoring and control of bioprocess performance. However, their implementation in real time requires further development of non-invasive analytical techniques and regulatory compliance. User-friendly interfaces are also needed to connect and contextualize information from different sensors through digitization [9, 28, 29].

3.3. Data formats and data storage

Due to digitalization and the increasing use of online-capable sensors, more and more data is being generated in bioproduction. Therefore, new strategies for data storage are being developed to cope with the large amounts of data and the increasing diversity and speed of data. These strategies include databases that are already used in other areas (e.g. corporate data), such as relational databases, non-relational databases, data warehouses and data lakes [30, 31]. An example for a non-relational database coupled with a bioprocess ontology to store and facilitate the reuse and extension of bioprocesses is NyctiDB [30].

The data generated by different sensors in bioproduction are different for each sensor class, e.g. spectral data [7], flow cytometry data [32] and turbidity sensors measuring biomass concentration [11]. The data format of stored bioprocess data is typically proprietary or in text or tabular form, despite the diverse nature of the measured parameters and data. It can be utilized as AI enablers for further research or training bioprocess simulators. Bioprocess simulators utilize biological data to parameterize process models, such as kinetic models of microbial growth and metabolism, and data on culture media, reactor design, and downstream processing. These models optimize bioprocesses, making them more efficient and cost-effective, ultimately improving the design, operation, and performance of bioprocesses. Open source simulators such as BioSTEAM [33] and the largest repository of bio models BioModels [34] enable greater accessibility and transparency in the scientific community and foster collaboration and innovation in the bioprocess industry.

3.4. Opportunities and Barriers

In bioproduction, the use of data by intelligent algorithms offers numerous opportunities for process optimization, such as increased efficiency, higher yields, less waste, and better product quality [2, 35]. These algorithms can analyze large and complex data sets generated by sensors and other monitoring tools and identify patterns and correlations not readily apparent to human analysts [35]. Using machine learning and other advanced analytics techniques, process parameters that have the greatest impact on product quality and yield can be identified, and changes can be recommended to improve overall process performance. In addition, data analysis can help understand the relationships between process parameters and cell-specific characteristics [36]. Intelligent algorithms can also help optimize production schedules, detect and correct process problems in real time, and predict equipment failures so that timely maintenance can be performed to avoid downtime, e.g. by regression analysis calculating and evaluating uncertainties [19, 36]. One of the main advantages of these algorithms is their ability to learn and adapt over time. As more data is collected and analyzed, the predictions and recommendations of these algorithms become more accurate, leading to even greater process optimization and efficiency [36].

The downside of ever-increasing amount of data, that is produces by research laboratories is, that it is becoming more complex and require further analysis. Lack of standardization and integrated storage of process design conditions and specifications within the biotech industry hinder model reuse within the scientific community, violating the FAIR principles (Findability, Accessibility, Interoperability, Reusability) [37]. Therefore, operational, and industry standard software systems are necessary for effective data management. Previous efforts in synthetic biology have made strides in this direction. However, the development of a bioprocess model's data storage system that aligns with the industry's needs and FAIR principles is still necessary [37].

Integrating automated and isolated workstations into a continuous workflow without compromising process efficiency remains a key challenge in bioproduction systems, especially due to lack of standardization. This has become more acute as bioprocesses become more complex and require higher levels of automation. Standardized communication protocols and graphical user interfaces are needed to achieve successful integration [9]. The scalability and flexibility of the middleware is also an important criterion for its acceptance by users. The next level of primary challenge is to integrate the enterprise control system with business systems, manufacturing execution systems, and manufacturing control systems. To overcome this challenge, monitoring and data acquisition platforms and ultimately the concept of digital twin must be introduced to enable control and monitoring in bioproduction systems [4, 38].

The lack of quantitative knowledge about cultivation behavior makes also process design and optimization difficult. Accurate mechanistic models for bioprocesses are also difficult to create and validate, requiring significant experimental effort and expertise [39]. Over parameterized models can lead to a lack of robustness and universality [30]. Bioproduction control faces challenges due to the complexity and nonlinearity of biological processes and the lack of reliable online sensors for key process variables. The complexity of metabolic networks makes it difficult to determine the best way to turn off specific genes, which can be addressed through the use of deep neural networks [26]. However, training such models is challenging due to the complexity of biochemical reactions and the lack of industrially deployable sensors [18]. Nonlinear behaviors in bioprocesses depend on various factors such as feed, temperature, and activity of microorganisms, making online monitoring essential for safe and high-quality product production. Researchers have proposed several methods to overcome this challenge, such as dynamic nonlinear monitoring methods that combine kernel principal component analysis (KPCA) and exponentially weighted moving average (EWMA) to better monitor bioprocesses [9, 20]. To address these challenges, researchers propose data-based methods such as online and historical databases [24]. However, searching and obtaining relevant information for process modeling can be difficult, and experiments remain a more accurate source of information, but also costly and time-consuming. Therefore, the development of an online database for process models is considered a possible solution to this problem [30]. Other challenges in bioproduction include limited knowledge about the interaction of adsorbents and bioprocesses, limited information about process variability in the early stages of process design and technology transfer, lack of industrially deployable sensors for online measurement of key process variables, difficulty in accessing some key process parameters, and conventional analytical methods that cannot generate real-time data [14, 40].

Furthermore, one primary objective of optimizing biological processes should not overlook the importance of sustainability [1, 41]. To achieve biointelligent production within planetary boundaries and in alignment with the environment, it is essential to strive for true sustainability and to design biological production accordingly Miehe et al. [1, 35].

4. Discussion

The outcomes of this paper provide an overview of recent research activities. The availability of data in bioproduction is increasing due to advances in sensor technology and the use of soft sensors. The use of data by intelligent algorithms in bioproduction offers numerous opportunities for process optimization, such as increased efficiency, higher yields, less waste, and better product quality. Intelligent algorithms can analyze large and complex data sets generated by sensors and other monitoring tools and identify patterns and correlations not readily apparent to human analysts. However, the downside of ever-increasing data is that it is becoming more complex and requires further analysis. Regulatory agencies are focusing on data integrity issues, and there is a lack of standardization and integrated storage of process design conditions and specifications within the biotech industry. The scalability and flexibility of the middleware are also important criteria for its acceptance by users.

Integrating automated and isolated workstations into a continuous workflow without compromising process efficiency remains a key challenge in bioproduction systems. Another challenge is the lack of quantitative knowledge about cultivation behavior, which makes process design and optimization difficult. Due to the complexity and nonlinearity of biological processes and the lack of reliable online sensors for key process variables, several challenges arise for bioproduction control. The complexity of metabolic networks makes it difficult to determine the best way to turn off specific genes, and the nonlinearity, variability, and complexity of biotechnological processes, especially fermentation processes, pose challenges for process monitoring and control.

5. Future Agenda

The results of the study indicate that the challenges in bioprocess monitoring and control require further research and analyses. Future research could focus on the development of integrated and standardized data storage systems, communication protocols and graphical user interfaces to achieve successful integration of automated and isolated workstations. In addition, further studies could focus on the development of online and historical databases and real-time sensors for key process variables to improve process monitoring and control. Finally, the development of mechanistic models for bioprocesses and the use of deep neural networks to learn patterns and create models that make sense of complex data could also be avenues for further research. In terms of analyzing new data storage technologies like data lakes and lakehouses for the storage of data in bioproduction, it would be important to consider how these technologies can be integrated with existing data storage and management systems. It would also be important to assess how the data model needs to be designed to ensure that data can be effectively used within these technologies for further analysis and decision-making.

The use of data in bioproduction offers numerous opportunities for optimization and increased efficiency, but also brings challenges related to standardization and compliance with data protection regulations. As organic production moves towards sustainability, the use of data can help achieve this goal.

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